

First Section.
The Elliptic Integrals.

§ 1. Definition of the elliptic integrals.

When the systematic presentation of the integral calculus has reached the point where algebraic integrals containing the square root of a function of the first or second degree are reduced to integrals of rational functions, the learner encounters there a barrier which he is not able to pass with the aids available up to that point. The effort to enlarge these aids, so as to make even the next class of integrals accessible to investigation, is, as it was historically the occasion for a more thorough study of the elliptic integrals and for the introduction of the elliptic functions, also the most natural and intelligible point of departure for one who is first to be introduced into the theory of these functions. Hence our next task is to become acquainted with the elliptic integrals and with their most important properties.

The definition of an elliptic integral from which we shall start is the following. Let $f(x)$ denote an integral rational function of the third or fourth degree with four distinct roots,

$$f(x) = a_0x^4 + 4a_1x^3 + 6a_2x^2 + 4a_3x + a_4, \quad (1)$$

where a_0 and a_1 do not vanish simultaneously; and let $\Phi(x, y)$ be an arbitrary integral or fractional rational function of the two arguments x, y . Then

$$\int \Phi(x, \sqrt{f(x)}) dx$$

is the general elliptic integral, and

$$\Phi(x, \sqrt{f(x)}) dx$$

is the general elliptic differential. But this general integral and differential can now be reduced to substantially simpler ones. First Φ can be put in the form

$$\frac{A + B\sqrt{f(x)}}{C + D\sqrt{f(x)}},$$

where A, B, C, D are integral rational functions of x ; and, by multiplying this fraction by $C - D\sqrt{f(x)}$, into the form

$$\Psi(x) + \frac{\Phi(x)}{\sqrt{f(x)}},$$

where $\Psi(x)$ and $\Phi(x)$ are integral or fractional rational functions of x , namely

$$\begin{aligned} \Psi(x) &= \frac{AC - BDf(x)}{C^2 - D^2f(x)}, \\ \Phi(x) &= \frac{(BC - AD)f(x)}{C^2 - D^2f(x)}. \end{aligned}$$

We now leave aside the rational integral

$$\int \Psi(x) dx,$$

which leads to logarithms and algebraic functions, and concern ourselves only with the elliptic integral

$$\int \frac{\Phi(x) dx}{\sqrt{f(x)}} \quad (2)$$

or the corresponding elliptic differential

$$\Phi(x) \frac{dx}{\sqrt{f(x)}}. \quad (3)$$

It is sometimes useful to consider this differential in homogeneous form. To this end we put for x the ratio of two variables $x : y$, and accordingly for dx

$$\frac{y dx - x dy}{y^2}.$$

If then

$$f(x, y) = a_0 x^4 + 4a_1 x^3 y + 6a_2 x^2 y^2 + 4a_3 x y^3 + a_4 y^4,$$

and $\Phi(x, y)$ is a homogeneous function of order 0, our elliptic differential takes the form

$$\Phi(x, y) \frac{y dx - x dy}{\sqrt{f(x, y)}}. \quad (4)$$

The problem that will occupy us first is twofold. First, by the introduction of new variables, the differential

$$\frac{dx}{\sqrt{f(x)}} \quad \text{or} \quad \frac{y dx - x dy}{\sqrt{f(x, y)}} \quad (5)$$

is to be transformed into another of the same form, but in which the function under the square-root sign is made as simple as possible, and in particular dependent on as small a number of parameters as possible (§§ 3, 4, 5). Second, the general differential (3) or (4) is to be decomposed into other similar differentials in which the function $\Phi(x)$ or $\Phi(x, y)$ is as simple as possible (§ 12).

The function $f(x)$ can be decomposed into linear factors (Vol. I, § 33). Accordingly we also put

$$f(x) = a_0(x - x_1)(x - x_2)(x - x_3)(x - x_4).$$

If we think of the quantities $x, x_1, x_2, \dots$ as abscissae of points laid off on a straight line from an arbitrary origin, positive on one side and negative on the other side (for example positive to the right, negative to the left), then the differences $x - x_1, x - x_2, x - x_3, x - x_4$ represent the distances of the point x from x_1, x_2, x_3, x_4 , counted positively when x lies on the positive side of $x_1, x_2, \dots$. In the same sense $x_2 - x_1$ is the distance of the point x_2 from x_1 , and so forth.

We shall denote the differences more briefly by $(xx_1), (xx_2), \dots, (xx_4), \dots$. We use this notation only for easier survey, and we do not exclude imaginary points of the line.

§ 2. Cross-ratios.

The simplest means of simplifying the elliptic differential rests on the linear transformation of the function $f(x)$ (Vol. I, § 67). Let $\alpha, \beta, \gamma, \delta$ denote four arbitrary constants whose determinant

$$\alpha\delta - \beta\gamma = r \quad (1)$$

is different from zero. We put

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta}, \quad (2)$$

and call this a linear substitution. If x and x' are interpreted, as in § 1, as points on two straight lines L and L' , then (2) determines a one-to-one correspondence of the points of L and L' , a mapping. To the values $x = \infty$ and $x' = \infty$ correspond the points at infinity of the lines, which we likewise regard as definite points. Thus the point $x' = \infty$ corresponds to the point $x = \alpha : \gamma$,

and the point $x = \infty$ to the point $x' = -\delta : \gamma$; and only when $\gamma = 0$, so that the substitution (2) is integral, do the points at infinity on L and L' correspond to one another.

The substitution (2) is not changed when the four transformation numbers are multiplied by the same factor. The determinant r is multiplied by the square of this factor, and therefore this factor can be so chosen that the determinant takes any prescribed value, for example the value 1.

If the substitution (2) is applied to any two points x_1, x_2 and the corresponding x'_1, x'_2 , then one obtains

$$(x_1 x_2) = \frac{r(x'_1 x'_2)}{(\gamma x'_1 + \delta)(\gamma x'_2 + \delta)},$$

and, if one adds a second pair of points x_3, x_4 and the corresponding x'_3, x'_4 ,

$$(x_1 x_2)(x_3 x_4) = \frac{r^2(x'_1 x'_2)(x'_3 x'_4)}{(\gamma x'_1 + \delta)(\gamma x'_2 + \delta)(\gamma x'_3 + \delta)(\gamma x'_4 + \delta)}.$$

Interchanging here x_2 with x_3 and forming the quotient, one obtains

$$\frac{(x_1 x_2)(x_3 x_4)}{(x_1 x_3)(x_2 x_4)} = \frac{(x'_1 x'_2)(x'_3 x'_4)}{(x'_1 x'_3)(x'_2 x'_4)}. \quad (3)$$

The expression on the left side is called the cross-ratio of the pair of points $x_1 x_4$ to the pair of points $x_2 x_3$, and thus this formula contains the theorem:

The cross-ratio of two pairs of points is invariant under simultaneous linear transformation of the four points.

Four points can be separated into two pairs in six ways. But if we put

$$a = (x_2 x_3)(x_1 x_4), \quad b = (x_3 x_1)(x_2 x_4), \quad c = (x_1 x_2)(x_3 x_4),$$

then the identity holds:

$$a + b + c = 0 \quad (4)$$

and we obtain the six cross-ratios

$$-\frac{c}{b}, \quad -\frac{a}{c}, \quad -\frac{b}{a}, \quad -\frac{b}{c}, \quad -\frac{c}{a}, \quad -\frac{a}{b}; \quad (5)$$

if we put the first of them, $-c : b$, equal to κ^2 , then from (4) we obtain the six:

$$\kappa^2, \quad 1 - \frac{1}{\kappa^2}, \quad \frac{1}{1 - \kappa^2}, \quad \frac{1}{\kappa^2}, \quad \frac{\kappa^2}{\kappa^2 - 1}, \quad 1 - \kappa^2. \quad (6)$$

Thus the six cross-ratios of the four points are expressed linearly through any one among them.

If x_1, x_2, x_3, x_4 are permuted among themselves, then a, b, c are permuted among themselves and change their signs, but in such a way that either all three signs are changed simultaneously or none is changed. For example the transpositions give

$$\begin{aligned} (14) \text{ and } (23) & \quad \begin{pmatrix} a & b & c \\ -a & -c & -b \end{pmatrix} \\ (24) \text{ and } (31) & \quad \begin{pmatrix} a & b & c \\ -c & -b & -a \end{pmatrix} \\ (34) \text{ and } (12) & \quad \begin{pmatrix} a & b & c \\ -b & -a & -c \end{pmatrix}. \end{aligned}$$

If x_1, x_2, x_3, x_4 are real, then κ^2 is a positive proper fraction if and only if κ^2 and $1 - \kappa^2$ are positive; that is, if either a and c are positive and b negative, or a and c negative and b positive.

This occurs when x_1, x_2, x_3, x_4 follow one another in this order increasingly in magnitude, and for all arrangements that arise from it by the permutations which leave b unchanged. These are the following eight:

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{array}$$

Thus, if one imagines x_1, x_2, x_3, x_4 placed in this order on the circumference of a circle, then one obtains a positive proper fraction κ^2 precisely when the x_i follow one another in magnitude in such a way that, beginning with an arbitrary one as the least, one then continues around the circle either to the right or to the left. We express this as follows:

In order that κ^2 be a positive proper fraction, the quantities x_1, x_2, x_3, x_4 must follow one another cyclically in order of magnitude.

§ 3. Linear transformation of the elliptic differential.

When the two points x_3 and x_4 approach one and the same point x , then x'_3 and x'_4 approach the corresponding point x' . We then put $(x_3x_4) : (x'_3x'_4) = dx : dx'$ and obtain from (3), § 2, the relation between the differentials dx, dx' :

$$\frac{(x_1x_2) dx}{(x_1x)(x_2x)} = \frac{(x'_1x'_2) dx'}{(x'_1x')(x'_2x')}. \quad (1)$$

Similarly, if in place of x_1, x_2 one puts two other points x_3, x_4 , one obtains

$$\frac{(x_3x_4) dx}{(x_3x)(x_4x)} = \frac{(x'_3x'_4) dx'}{(x'_3x')(x'_4x')}, \quad (2)$$

and, multiplying and taking the square root,

$$\frac{\sqrt{(x_1x_2)(x_3x_4)} dx}{\sqrt{(x_1x)(x_2x)(x_3x)(x_4x)}} = \frac{\sqrt{(x'_1x'_2)(x'_3x'_4)} dx'}{\sqrt{(x'_1x')(x'_2x')(x'_3x')(x'_4x')}} \quad (3)$$

and therefore a transformation of the elliptic differential by a linear substitution:

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta}.$$

This substitution is completely determined if to three arbitrary points x_1, x_2, x_3 the corresponding values x'_1, x'_2, x'_3 are arbitrarily assigned; it is obtained from the equality of the cross-ratios

$$\frac{(x_1x_2)(x_3x)}{(x_1x_3)(x_2x)} = \frac{(x'_1x'_2)(x'_3x')}{(x'_1x'_3)(x'_2x')}, \quad (4)$$

by solving for x , and the fourth value x'_4 belonging to x_4 is obtained from

$$\frac{(x_1x_2)(x_3x_4)}{(x_1x_3)(x_2x_4)} = \frac{(x'_1x'_2)(x'_3x'_4)}{(x'_1x'_3)(x'_2x'_4)}. \quad (5)$$

If the function $f(x)$ is given in the elliptic differential

$$du = \frac{dx}{\sqrt{f(x)}},$$

then x_1, x_2, x_3, x_4 are thereby also given, and by means of the linear transformation (3) one can bring the differential into another form in which three of the quantities x'_1, x'_2, x'_3, x'_4 have arbitrarily assigned values. One obtains the normal form if one assumes

$$x' = z, \quad x'_1 = 0, \quad x'_2 = 1, \quad x'_3 = \frac{1}{\kappa^2}, \quad x'_4 = \infty.$$

The quantity κ is called by Legendre the modulus of the elliptic differential, and $\sqrt{1 - \kappa^2} = \kappa'$ the complementary modulus. Thus, if the points are made to correspond in the following way,

$$\begin{array}{cccccc} z, & 0, & 1, & \frac{1}{\kappa^2}, & \infty, \\ x, & x_1, & x_2, & x_3, & x_4, \end{array}$$

then from the equality of cross-ratios one easily obtains the relations

$$\begin{aligned} z &= \frac{(xx_1)(x_2x_4)}{(xx_4)(x_2x_1)}, \\ 1 - z &= \frac{(xx_2)(x_1x_4)}{(xx_4)(x_1x_2)}, \\ 1 - \kappa^2 z &= \frac{(xx_3)(x_1x_4)}{(xx_4)(x_1x_3)}, \end{aligned} \tag{6}$$

$$dz = \frac{(x_4x_1)(x_4x_2)}{(x_2x_1)} \frac{dx}{(xx_4)^2}, \tag{7}$$

$$\kappa^2 = \frac{(x_3x_4)(x_1x_2)}{(x_1x_3)(x_2x_4)}, \quad \kappa'^2 = \frac{(x_2x_3)(x_1x_4)}{(x_2x_4)(x_1x_3)}, \tag{8}$$

and from (3)

$$\frac{\sqrt{(x_3x_1)(x_4x_2)} dx}{\sqrt{-(x_1x)(x_2x)(x_3x)(x_4x)}} = \frac{dz}{\sqrt{z(1-z)(1-\kappa^2 z)}}. \tag{9}$$

By § 2, κ^2 is a positive proper fraction when x_1, x_2, x_3, x_4 are real and follow one another cyclically in order of magnitude. This gives eight different transformations of this kind, and among them one can choose two for each case in which the interval $z = 0$ to $z = 1$ corresponds to one of the prescribed intervals $(x_1x_2), (x_2x_3), (x_3x_4), (x_4x_1)$; for one of these transformations increasing z corresponds to increasing x , and for the other to decreasing x (with passage from $+\infty$ to $-\infty$). In the first case the square roots in (9) have the same sign on both sides, in the second the opposite sign.

If one does not insist that κ^2 be a positive proper fraction, then one may permute the points x_1, x_2, x_3, x_4 in all possible ways and obtains 24 distinct linear transformations into the normal form, of which any four give the same κ^2 ; in all there are six different moduli, derived from one another according to § 2, (6).

The totality of these transformations is most easily surveyed if one assumes that the integral to be transformed already has the normal form

$$\frac{dx}{\sqrt{x(1-x)(1-\lambda^2 x)}};$$

then in the formulae (6) to (9) one has only to replace x_1, x_2, x_3, x_4 in all possible ways by $0, 1, 1 : \lambda^2, \infty$. As examples we take the following correspondences:

	x_1	x_2	x_3	x_4
1)	0	∞	$\frac{1}{\lambda^2}$	1
2)	0	$\frac{1}{\lambda^2}$	1	∞
3)	1	0	∞	$\frac{1}{\lambda^2}$

From (6) and (8) one then obtains

$$\begin{aligned} 1) \quad z &= -\frac{x}{1-x}, \quad \varkappa^2 = 1 - \lambda^2 = \lambda'^2, \\ 2) \quad z &= \lambda^2 x, \quad \varkappa^2 = \frac{1}{\lambda^2}, \\ 3) \quad z &= \frac{1-x}{1-\lambda^2 x}, \quad \varkappa^2 = \lambda^2, \end{aligned}$$

and for the differential

$$\frac{dz}{\sqrt{z(1-z)(1-\varkappa^2 z)}}$$

one obtains, in the three cases,

$$\begin{aligned} 1) \quad & \frac{dx}{\sqrt{-x(1-x)(1-\lambda^2 x)}}, \\ 2) \quad & \frac{\lambda dx}{\sqrt{x(1-x)(1-\lambda^2 x)}}, \\ 3) \quad & \frac{dx}{\sqrt{x(1-x)(1-\lambda^2 x)}}. \end{aligned}$$

§ 4. Legendre's normal form.

If the function $f(x)$ which stands under the root sign in the elliptic differential has real coefficients, three cases are possible:

1. $f(x)$ has four real roots x_1, x_2, x_3, x_4 ;
2. $f(x)$ has two real roots x_1, x_2 and a pair of conjugate imaginary roots x_3, x_4 ;
3. $f(x)$ has two pairs of conjugate imaginary roots x_1, x_2 and x_3, x_4 .

In all three cases the elliptic differential can be brought by a real linear transformation to the form

$$\frac{dz}{\sqrt{(z^2 - \alpha)(z^2 - \beta)}}, \tag{1}$$

where α and β are real, positive or negative, constants.

We determine the linear dependence between the variables x and z so that the following values correspond:

$$\begin{array}{ccccc} x, & x_1, & x_2, & x_3, & x_4, \\ z, & \sqrt{\alpha}, & -\sqrt{\alpha}, & \sqrt{\beta}, & -\sqrt{\beta}, \end{array} \tag{2}$$

and obtain a substitution of the form

$$h \frac{x - x_1}{x - x_2} = \frac{z - \sqrt{\alpha}}{z + \sqrt{\alpha}}. \tag{3}$$

To determine h , we put $x = x_3$, $x = x_4$, and correspondingly $z = \sqrt{\beta}$, $z = -\sqrt{\beta}$; this gives

$$h \frac{(x_3x_1)}{(x_3x_2)} = \frac{\sqrt{\beta} - \sqrt{\alpha}}{\sqrt{\beta} + \sqrt{\alpha}}, \quad h \frac{(x_4x_1)}{(x_4x_2)} = \frac{\sqrt{\beta} + \sqrt{\alpha}}{\sqrt{\beta} - \sqrt{\alpha}}.$$

By multiplication and division it follows that

$$h = \sqrt{\frac{(x_3x_2)(x_4x_2)}{(x_3x_1)(x_4x_1)}}, \quad \frac{\sqrt{\alpha} - \sqrt{\beta}}{\sqrt{\alpha} + \sqrt{\beta}} = \sqrt{\frac{(x_3x_1)(x_4x_2)}{(x_3x_2)(x_4x_1)}}. \quad (4)$$

If, as in § 2, we put

$$\begin{aligned} a &= (x_2x_3)(x_1x_4), \\ b &= (x_3x_1)(x_2x_4), \\ c &= (x_1x_2)(x_3x_4), \\ a + b + c &= 0, \end{aligned} \quad (5)$$

then, since only the ratio of α to β matters, we can satisfy the last of equations (4) by putting

$$\sqrt{\alpha} = \sqrt{\pm a} + \sqrt{\mp b}, \quad \sqrt{\beta} = \sqrt{\pm a} - \sqrt{\mp b}, \quad (6)$$

and from (3) one obtains for z the expression

$$z = \sqrt{\alpha} \frac{(xx_2)\sqrt{(x_3x_1)(x_4x_1)} + (xx_1)\sqrt{(x_3x_2)(x_4x_2)}}{(xx_2)\sqrt{(x_3x_1)(x_4x_1)} - (xx_1)\sqrt{(x_3x_2)(x_4x_2)}}. \quad (7)$$

Putting beside (3) the equation following from it,

$$h' \frac{x - x_3}{x - x_4} = \frac{z - \sqrt{\beta}}{z + \sqrt{\beta}},$$

one obtains by logarithmic differentiation

$$\frac{(x_1x_2)dx}{(xx_1)(xx_2)} = \frac{2\sqrt{\alpha}dz}{z^2 - \alpha}, \quad \frac{(x_3x_4)dx}{(xx_3)(xx_4)} = \frac{2\sqrt{\beta}dz}{z^2 - \beta},$$

and hence, by multiplication and with regard to (5) and (6),

$$\frac{dx}{\sqrt{\pm(xx_1)(xx_2)(xx_3)(xx_4)}} = \frac{2dz}{\sqrt{(z^2 - \alpha)(z^2 - \beta)}}. \quad (8)$$

If now the four roots of $f(x)$ are real, then, as we saw in § 2, there are eight ways to assign these roots to the signs x_1, x_2, x_3, x_4 so that a and b have opposite signs; and if therefore $\pm a, \mp b$ are positive, $\sqrt{\alpha}, \sqrt{\beta}$ become real, hence α, β positive, and then by (7) z also becomes real.

Second, if x_1, x_2 are real and x_3, x_4 conjugate imaginary, then a and $-b$ are conjugate imaginary; hence, if the signs of the square roots $\sqrt{\pm a}, \sqrt{\mp b}$ are suitably chosen, $\sqrt{\alpha}$ is real and $\sqrt{\beta}$ purely imaginary, so that α is positive, β negative, and z real, since $(x_3x_1)(x_4x_1)$ and $(x_3x_2)(x_4x_2)$ are positive as products of conjugate imaginary quantities.

Finally, if x_1, x_2 and x_3, x_4 are two conjugate imaginary pairs, then a and $-b$ are both positive. If therefore in (6) we take the lower signs, then $\sqrt{\alpha}, \sqrt{\beta}$ are purely imaginary, hence α and β are negative, and from (7) z is a real expression.

Putting next

$$z^2 = y, \quad (9)$$

one obtains

$$\frac{2dz}{\sqrt{(z^2 - \alpha)(z^2 - \beta)}} = \frac{dy}{\sqrt{y(y - \alpha)(y - \beta)}} \quad (10)$$

and hence, by the quadratic substitution (9), an elliptic differential in which the expression under the root sign is a function of the third degree with real roots. By § 3 this can be brought by a linear substitution to the normal form

$$\frac{d\xi}{\sqrt{\xi(1-\xi)(1-\kappa^2\xi)}},$$

and in it ξ and κ can be taken as positive proper fractions.

Legendre's normal form follows from this by putting

$$\xi = \sin^2 \varphi, \quad d\xi = 2 \sin \varphi \cos \varphi d\varphi;$$

then

$$\frac{d\xi}{\sqrt{\xi(1-\xi)(1-\kappa^2\xi)}} = \frac{2d\varphi}{\sqrt{1-\kappa^2 \sin^2 \varphi}}. \quad (11)$$

§ 5. Weierstrass's normal form.

Another normal form of the elliptic differential, the one on which Weierstrass based his investigations, is

$$du = \frac{dz}{\sqrt{4z^3 - g_2z - g_3}}, \quad (1)$$

where g_2, g_3 are constants, called the first and second invariants of the differential. The general elliptic differential

$$\frac{dx}{\sqrt{f(x)}}, \quad (2)$$

where

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4, \quad (3)$$

can be brought by a linear transformation to Weierstrass's normal form if the roots x_1, x_2, x_3, x_4 of $f(x)$ are known. This is most simply done as follows.

Let the roots of the cubic function

$$\varphi(z) = 4z^3 - g_2z - g_3$$

be denoted by e_1, e_2, e_3 . Then

$$\varphi(z) = 4(z - e_1)(z - e_2)(z - e_3)$$

and $e_1 + e_2 + e_3 = 0$.

We let the values of x and z correspond to one another as follows:

$$\begin{array}{cccccc} x, & x_1, & x_2, & x_3, & x_4, \\ z, & \infty, & e_1, & e_2, & e_3, \end{array}$$

and it follows, if m denotes a constant factor that may be chosen arbitrarily, that

$$\begin{aligned} z - e_1 &= \frac{m(xx_2)}{(x_1x_2)(xx_1)} = m \left(\frac{1}{(x_1x_2)} - \frac{1}{(x_1x)} \right), \\ z - e_2 &= \frac{m(xx_3)}{(x_1x_3)(xx_1)} = m \left(\frac{1}{(x_1x_3)} - \frac{1}{(x_1x)} \right), \\ z - e_3 &= \frac{m(xx_4)}{(x_1x_4)(xx_1)} = m \left(\frac{1}{(x_1x_4)} - \frac{1}{(x_1x)} \right). \end{aligned} \quad (4)$$

The factor m must be the same in all three equations in order that the differences $(z - e_3) - (z - e_2) = e_2 - e_3$, and so on, be independent of x .

Forming these differences and putting, as in Algebra, Vol. I, § 70,

$$(x_1x_2)(x_3x_4) = U, \quad (x_1x_3)(x_4x_2) = V, \quad (x_1x_4)(x_2x_3) = W,$$

one obtains

$$e_2 - e_3 = \frac{-mU}{(x_1x_2)(x_1x_3)(x_1x_4)}.$$

Introducing a new arbitrary factor μ by putting

$$m = 3\mu a_0(x_1x_2)(x_1x_3)(x_1x_4), \quad (5)$$

one obtains

$$e_2 - e_3 = -3\mu a_0 U, \quad e_3 - e_1 = -3\mu a_0 V, \quad e_1 - e_2 = -3\mu a_0 W,$$

and, since $e_1 + e_2 + e_3 = 0$,

$$e_1 = \mu a_0(V - W), \quad e_2 = \mu a_0(W - U), \quad e_3 = \mu a_0(U - V).$$

The quantities y_1, y_2, y_3 introduced in Algebra I, § 70, are therefore equal to $e_1 : \mu, e_2 : \mu, e_3 : \mu$, and by formula (11) there one obtains for e_1, e_2, e_3 the cubic equation

$$z^3 - 3A\mu^2z + \mu^3B = 0.$$

Thus, if $\varphi(z) = z^3 - g_2z - g_3$ and if

$$\varphi(z) = 4(z - e_1)(z - e_2)(z - e_3) = 4z^3 - g_2z - g_3$$

is to hold, then

$$g_2 = 12\mu^2A, \quad g_3 = -4\mu^3B. \quad (6)$$

Here

$$\begin{aligned} A &= a_2^2 - 3a_1a_3 + 12a_0a_4, \\ B &= 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3 \end{aligned} \quad (7)$$

are the first and second invariants of the biquadratic form $f(x)$.

Logarithmic differentiation of the first two equations (4) gives

$$\frac{dz^2}{(z - e_1)(z - e_2)} = \frac{dx^2(x_1x_2)(x_1x_3)}{(xx_1)^2(xx_2)(xx_3)},$$

and with the help of the last equation (4), by (5),

$$\frac{dz}{\sqrt{4z^3 - g_2z - g_3}} = \frac{dx}{\sqrt{12\mu f(x)}}. \quad (8)$$

If we wish to apply these results to the case where $f(x)$ has the normal form

$$f(x) = x(1 - x)(1 - \varkappa^2x) = x - x^2(1 + \varkappa^2) + \varkappa^2x^3,$$

and the point $z = \infty$ corresponds to the point $x = 0$, then we let a_0 tend to zero and x_4 to infinity, but let the product a_0x_4 tend to a finite value, which we may take to be 1.

Then

$$a_0x_4 = 1, \quad a_0 = 0, \quad a_1 = \varkappa^2,$$

$$a_2 = -(1 + \varkappa^2), \quad a_3 = 1, \quad a_4 = 0,$$

and consequently

$$\begin{aligned} A &= (1 + \varkappa^2)^2 - 3\varkappa^2 = 1 - \varkappa^2 - \varkappa^4 = 1 - \varkappa^2 \varkappa'^2, \\ B &= -(1 + \varkappa^2)\{2(1 + \varkappa^2)^2 - 9\varkappa^2\} \\ &= -(1 + \varkappa^2)(2 - \varkappa^2)(1 - 2\varkappa^2) \\ &= (1 + \varkappa^2)(1 + \varkappa'^2)(\varkappa^2 - \varkappa'^2) \\ &= (2 + \varkappa^2 \varkappa'^2)(\varkappa^2 - \varkappa'^2), \end{aligned}$$

where $\varkappa'^2 = 1 - \varkappa^2$.

Thus, if we also add the discriminant

$$\Delta = 16 \cdot 27 \mu^6 (4A^3 - B^2),$$

then

$$\begin{aligned} g_2 &= 12\mu^2(1 - \varkappa^2 \varkappa'^2), \\ g_3 &= -4\mu^3(2 + \varkappa^2 \varkappa'^2)(\varkappa^2 - \varkappa'^2), \\ \Delta &= g_2^3 - 27g_3^2 = 27^2 \cdot 16 \mu^6 \varkappa^4 \varkappa'^4, \end{aligned} \tag{9}$$

and from (6) one obtains the transformation

$$\frac{dz}{\sqrt{4z^3 - g_2z - g_3}} = \frac{dx}{\sqrt{12\mu x(1-x)(1-\varkappa^2 x)}}. \tag{10}$$

To find the substitution, we put $x_1 = 0$, $x_2 = 1$, $x_3 = 1 : \varkappa^2$, $x_4 = \infty$, $a_0 x_4 = 1$, and obtain

$$a_0 U = 1, \quad a_0 V = -\frac{1}{\varkappa^2}, \quad a_0 W = \frac{\varkappa'^2}{\varkappa^2},$$

so that

$$\begin{aligned} e_1 &= \mu \frac{\varkappa^2 - 2}{\varkappa^2}, \quad e_2 = \mu \frac{1 - 2\varkappa^2}{\varkappa^2}, \quad e_3 = \mu \frac{1 + \varkappa^2}{\varkappa^2}, \\ z &= \frac{3\mu}{\varkappa^2} \left(\frac{\varkappa^2 + 1}{3} - \frac{1}{x} \right). \end{aligned} \tag{11}$$

In this transformation of the elliptic differential into Weierstrass's normal form, the decomposition of the function $f(x)$ into its linear factors, and hence the solution of the biquadratic equation $f(x) = 0$, is presupposed. Another transformation, not linear, to be sure, which reaches the same goal without this presupposition, was given by Hermite (Crelle's Journal, vol. 52). To present it, we use the homogeneous form of the elliptic differential

$$\frac{y dx - x dy}{\sqrt{f(x, y)}}, \tag{12}$$

where

$$f(x, y) = a_0 x^4 + a_1 x^3 y + a_2 x^2 y^2 + a_3 x y^3 + a_4 y^4. \tag{13}$$

Besides the two invariants A, B , this biquadratic form has two further covariants:

$$H = \frac{1}{3} \left[\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \right], \quad T = \frac{1}{12} \left(\frac{\partial f}{\partial x} \frac{\partial H}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial H}{\partial x} \right), \tag{14}$$

where H and T are forms of the fourth and sixth degrees whose coefficients are composed rationally, and with integral numerical coefficients, from the coefficients of f . Between these forms there is the further relation

$$H^3 - 48AHf^2 - 64Bf^3 = -27T^2 \tag{15}$$

(Vol. I, §§ 70, 72).

But by Euler's theorem on homogeneous functions,

$$4f = \frac{\partial f}{\partial x}x + \frac{\partial f}{\partial y}y, \quad df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy,$$

$$4H = \frac{\partial H}{\partial x}x + \frac{\partial H}{\partial y}y, \quad dH = \frac{\partial H}{\partial x}dx + \frac{\partial H}{\partial y}dy,$$

whence

$$f dH - H df = -3T(y dx - x dy),$$

and if, according to (15), one puts

$$3\sqrt{3}T = \sqrt{-H^3 + 48Af^2H + 64Bf^3}, \quad (16)$$

then

$$\frac{y dx - x dy}{\sqrt{f}} = \frac{-\sqrt{3}(f dH - H df)}{\sqrt{-(H^3 - 48Af^2H - 64Bf^3)}f}.$$

If one now makes the substitution

$$z = -\frac{\mu H}{4f} \quad (17)$$

with an undetermined factor μ , then one obtains

$$\frac{y dx - x dy}{\sqrt{3\mu f}} = \frac{dz}{\sqrt{4z^3 - g_2z - g_3}}, \quad (18)$$

where, as before,

$$g_2 = 12\mu^2A, \quad g_3 = -4\mu^3B. \quad (19)$$

This transformation can be applied to a differential which already has the normal form. If one puts

$$f(x, y) = 4x^3y - g_2xy^3 - g_3y^4,$$

then one has to replace

$$\begin{array}{ccccc} a_0 & a_1 & a_2 & a_3 & a_4 \\ 0 & 4 & 0 & -g_2 & -g_3 \end{array}$$

and from (7) one obtains

$$A = 12g_2, \quad B = -27 \cdot 16 g_3.$$

If therefore one puts $\mu = \frac{1}{12}$, then equations (19) pass into the identities $g_2 = g_2$, $g_3 = g_3$.

If one then puts $y = 1$, transformation (18) gives

$$\frac{2dx}{\sqrt{4x^3 - g_2x - g_3}} = \frac{dz}{\sqrt{4z^3 - g_2z - g_3}}. \quad (20)$$

By (16), this transformation is mediated by

$$z = -\frac{H}{48f}.$$

But from (14) one obtains

$$H = -48 \left[\left(x^2 + \frac{1}{4}g_2 \right)^2 + 2g_3x \right],$$

and hence

$$z = \frac{\left(x^2 + \frac{1}{4}g_2\right)^2 + 2g_3x}{4x^3 - g_2x - g_3}. \quad (21)$$

Furthermore, from (14) one obtains

$$T = 64x^6 - 80g_2x^4 - 320g_3x^3 - 20g_2^2x^2 - 16g_2g_3x + g_2^3 - 32g_3^2,$$

and then, by (16),

$$4\left(\sqrt{4x^3 - g_2x - g_3}\right)^3 \sqrt{4z^3 - g_2z - g_3} = T. \quad (22)$$

Equations (20), (21), (22) carry out the multiplication of the elliptic differential by 2. In this way not only is z represented as a single-valued rational function of x , but one square root is also represented uniquely by the other, that is, to a point x there is assigned a point z uniquely.

§ 6. Elliptic curves.

Every algebraic dependence between two variables x, y is expressed by an equation of the form

$$F(x, y) = 0, \quad (1)$$

where $F(x, y)$ denotes an integral function of the two variables x, y . If one regards x and y as Cartesian coordinates of a point in the plane, then (1) is the equation of a curve of the n th degree when F is of the n th dimension with respect to x and y together. This curve is then the geometric image of the algebraic dependence, although imaginary points too must be taken into account.

If $\Phi(x, y)$ is an integral or fractional rational function of x and y , and y depends on x through equation (1), then

$$\Phi(x, y) dx, \quad \int \Phi(x, y) dx \quad (2)$$

are the algebraic differentials and integrals belonging to the curve F .

For example, if $f(x)$ is a function of the third or fourth degree in x , then

$$y^2 - f(x) = 0$$

is the equation of a curve of the third or fourth degree to which belong the elliptic differentials and integrals affected with the irrationality $\sqrt{f(x)}$.

We shall here call all curves whose differentials and integrals are reducible to elliptic ones elliptic curves. As the example shows, curves of the third and fourth degrees occur among them.

Conic sections do not belong to the elliptic curves, because, if an equation of the second degree holds between x and y , then x and y can be represented as rational functions of a parameter t , whereby the algebraic differential can be reduced to a rational one in t . Such curves are called rational curves.

We shall see that all curves of the third degree without a double point or cusp belong to the elliptic ones. Curves of higher degree can belong here only when they have a certain number of singular points. This is a fundamental chapter in the general theory of algebraic functions and does not belong to the plan of this work.¹

¹The most important works concerning this subject are: Riemann, Theorie der Abelschen Funktionen [Crelle's Journal, vol. 54 (1857)]. Mathematische Werke, 2nd ed., pp. 88, 467. Nachträge, edited by Noether and Wirtinger (1902). Aronhold, Monatsberichte der Berliner Akademie of 25 April 1861. Clebsch, Über die Anwendung der Abelschen Funktionen in der Geometrie (Crelle's Journal, vol. 63). Clebsch, Über diejenigen ebenen Kurven, deren Koordinaten rationale Funktionen eines Parameters sind (ibid., vol. 64). Clebsch, Über diejenigen Kurven, deren Koordinaten sich als elliptische Funktionen eines Parameters darstellen lassen (ibid., vol. 64). Brill, Über diejenigen Kurven, deren Koordinaten sich als hyperelliptische Funktionen eines Parameters darstellen lassen (ibid., vol. 65). Clebsch and Gordan, Theorie der Abelschen Funktionen (Teubner, Leipzig 1866).

For the investigation of algebraic differentials from this point of view, the introduction of homogeneous variables is expedient. We put $x = x_1 : x_3$, $y = x_2 : x_3$, $x_3^n F(x, y) = f(x_1, x_2, x_3)$; then $f(x_1, x_2, x_3)$ is a homogeneous function of order n of the three variables x_1, x_2, x_3 , and

$$f(x_1, x_2, x_3) = 0 \quad (3)$$

is the equation of a curve of order n in homogeneous coordinates. We have

$$dx = \frac{x_3 dx_1 - x_1 dx_3}{x_3^2}$$

and

$$d\Omega = \Phi(x, y)dx = \frac{1}{x_3^2} \Phi\left(\frac{x_1}{x_3}, \frac{x_2}{x_3}\right) (x_3 dx_1 - x_1 dx_3). \quad (4)$$

But by Euler's theorem on homogeneous functions, if f_1, f_2, f_3 denote the partial derivatives of f with respect to x_1, x_2, x_3 , then

$$f_1 x_1 + f_2 x_2 + f_3 x_3 = 0, \quad f_1 dx_1 + f_2 dx_2 + f_3 dx_3 = 0.$$

Therefore, if ρ denotes a factor of proportionality,

$$f_1 = \rho(x_2 dx_3 - x_3 dx_2), \quad f_2 = \rho(x_3 dx_1 - x_1 dx_3), \quad f_3 = \rho(x_1 dx_2 - x_2 dx_1),$$

and if c_1, c_2, c_3 denote wholly arbitrary quantities, for example constants,

$$c_1 f_1 + c_2 f_2 + c_3 f_3 = \rho \sum \pm c_1 x_2 dx_3,$$

where $\sum \pm c_1 x_2 dx_3$ denotes, in the usual way, the determinant

$$\begin{vmatrix} c_1 & c_2 & c_3 \\ x_1 & x_2 & x_3 \\ dx_1 & dx_2 & dx_3 \end{vmatrix}.$$

It follows that

$$x_3 dx_1 - x_1 dx_3 = \frac{f_2 \sum \pm c_1 x_2 dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3},$$

and, if one puts

$$\frac{f_2}{x_3^2} \Phi\left(\frac{x_1}{x_3}, \frac{x_2}{x_3}\right) = \Psi,$$

there results from (4) the expression for the most general algebraic differential belonging to the curve f :

$$d\Omega = \frac{\Psi \sum \pm c_1 x_2 dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3}, \quad (5)$$

where Ψ is an integral or fractional homogeneous function of order $n-3$. The arbitrary quantities c_1, c_2, c_3 occur in this expression only in appearance. In reality it is wholly independent of them.

We again consider the case $n = 3$. Then the simplest case is the one in which Ψ is a constant, and (5) passes into the elliptic differential of the first kind:

$$du = \frac{\sum \pm c_1 x_2 dx_3}{c_1 f_1 + c_2 f_2 + c_3 f_3}. \quad (6)$$

We shall now transform this, with the aid of the covariants of the ternary form of the third degree, to Weierstrass's form.

In Vol. II, § 107, we became acquainted with the fundamental covariants of the cubic form. Apart from f itself, these were

$$\Delta = \frac{1}{6^3} \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix}, \quad J = \frac{1}{36} \begin{vmatrix} f_{11} & f_{12} & f_{13} & \Delta_1 \\ f_{21} & f_{22} & f_{23} & \Delta_2 \\ f_{31} & f_{32} & f_{33} & \Delta_3 \\ \Delta_1 & \Delta_2 & \Delta_3 & 0 \end{vmatrix},$$

$$K = \frac{1}{9} \begin{vmatrix} f_1 & \Delta_1 & J_1 \\ f_2 & \Delta_2 & J_2 \\ f_3 & \Delta_3 & J_3 \end{vmatrix},$$

where the indices 1, 2, 3 denote differentiation with respect to the variables x_1, x_2, x_3 .

From the multiplication theorem for determinants it follows, since f and df are zero, that

$$\begin{vmatrix} c_1 & c_2 & c_3 \\ x_1 & x_2 & x_3 \\ dx_1 & dx_2 & dx_3 \end{vmatrix} \begin{vmatrix} f_1 & f_2 & f_3 \\ \Delta_1 & \Delta_2 & \Delta_3 \\ J_1 & J_2 & J_3 \end{vmatrix} = 3 \sum c_i f_i (\Delta dJ - 2J d\Delta),$$

and hence

$$du = 3 \frac{\Delta dJ - 2J d\Delta}{K}. \quad (7)$$

With the help of the covariants we have reduced the form $f(x)$ to the canonical form

$$\varphi(y) = y_1^3 + y_2^3 + y_3^3 + 6my_1y_2y_3.$$

If r denotes the substitution determinant, we put

$$P = \frac{(2 + m^3)m^3f^2 + (2 - 5m^3)mr^2f\Delta + 3m^2r^4\Delta^2 - r^6J}{(1 + 8m^3)^2},$$

$$Q = \frac{(1 + 2m^3)f - 6mr^2\Delta}{1 + 8m^3}, \quad R = \frac{m^2f + r^2\Delta}{1 + 8m^3},$$

and obtained y_1^3, y_2^3, y_3^3 as the roots of the cubic equation

$$u^3 - Qu^2 + Pu - R^3 = 0,$$

and the discriminant of this cubic equation is

$$D_1 = \frac{r^{18}K^2}{(1 + 8m^3)^6}.$$

If one expresses this discriminant through P, Q, R , then K^2 is represented rationally through f, Δ, J . At the place mentioned in the Algebra we did not give this expression explicitly. Now, however, we must form it, even though only under the presupposition $f = 0$, that is, only for the points of the curve. But [by Vol. I, § 50, (10)]

$$D_1 = P^2Q^2 + 18PQR^3 - 4P^3 - 4Q^3R^3 - 27R^6.$$

In this one must set

$$P = \frac{3m^2r^4\Delta^2 - r^6J}{(1 + 8m^3)^2}, \quad Q = \frac{-6mr^2\Delta}{1 + 8m^3}, \quad R = \frac{r^2\Delta}{1 + 8m^3},$$

and by a not difficult calculation one obtains, denoting by S and T the two invariants of the curve of the third order (Vol. II, § 108),

$$(1 + 8m^3)^6 D_1 = r^{18}(4J^3 + 108SJ\Delta^4 - 27T\Delta^6),$$

and consequently

$$K^2 = 4J^3 + 108SJ\Delta^4 - 27T\Delta^6. \quad (8)$$

This equation, however, is not identical, but is satisfied only under the presupposition $f = 0$. The complete expression of K^2 through J, Δ, f becomes much more complicated.

If one puts

$$\frac{J}{\Delta^2} = 3z, \quad dz = \frac{1}{3} \frac{\Delta dJ - 2J d\Delta}{\Delta^3}, \quad (9)$$

then from (8)

$$K^2 = 27\Delta^6(4z^3 + 12Sz - T),$$

and from (7)

$$du = \frac{1}{\sqrt{3}} \frac{dz}{\sqrt{4z^3 + 12Sz - T}}, \quad (10)$$

and this is Weierstrass's normal form for $g_2 = -12S$, $g_3 = T$.